Yousef Fahim Amin

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PROFESSIONAL SUMMARY

Computer Science student specializing in Artificial Intelligence and Software Development. Skilled in Python, TensorFlow, OpenCV, Flutter, and Arduino, with experience in machine learning, computer vision, and mobile app development. Interested in Computer Vision, Intelligent Systems, and applied AI research.

SKILLS

- **Languages:** Python, C++, Java, Dart
- **AI/ML:** Scikit-learn, TensorFlow/Keras, OpenCV, MediaPipe, Pandas, NumPy
- **Embedded Systems:** Arduino, Embedded C/C++, GPIO, PWM, UART
- **Development:** Flutter, Firebase, Streamlit
- **Tools:** Git, GitHub, Arduino IDE

PROJECTS

Hand Gesture Volume Control | GitHub

- Developed a real-time hand gesture volume controller using Python, OpenCV, MediaPipe, and PyCAW.
- Implemented hand landmark tracking for smooth gesture-based system control.

Arduino Projects Collection | GitHub

- Engineered embedded systems projects using Arduino and Embedded C/C++ including a Simon Says game, PWM light dimmer, and plant monitoring system.
- Integrated sensors, UART communication, and analog/digital I/O for real-time hardware interaction.

Comprehensive ML Pipeline – Heart Disease Prediction | GitHub

- Designed an end-to-end ML pipeline including preprocessing, feature engineering, PCA, and model training.
- Trained and deployed Logistic Regression, Random Forest, and SVM models using Streamlit.

FitTracker Pro – Gym & Fitness Mobile App | GitHub

- Developed a cross-platform fitness tracking app using Flutter and Firebase.
- Implemented workout management, authentication, analytics dashboards, and profile features.

Digit Recognition System | GitHub

- Built a handwritten digit recognition system using machine learning models on the scikit-learn dataset.
- Trained SVM and Decision Tree models after preprocessing and feature preparation.

EDUCATION

B.Sc. in Computer Science — Modern Academy, Maadi (Expected 2027).
Distinctions in Numerical Analysis, Calculus, and Linear Algebra.

EXPERIENCE

- **AI & ML Trainee — Sprints × Microsoft Summer Camp:** 3-month online training in ML workflows, supervised/unsupervised learning, PCA, SVM, and Random Forest. | GitHub | Certificate
- **AI & ML Trainee — IT Gate Academy:** Built SVM and Decision Tree models for digit classification with 98.89% SVM accuracy. | GitHub
- **Python Programming Trainee — Ex-Fresher:** 6-month onsite/online Python training covering OOP, data structures, exception handling, and team-based projects. | Certificate

Relevant Coursework:

Linear Algebra, Calculus, Numerical Analysis, Data Structures, Algorithms, Probability

Research Interests:

Artificial Intelligence, Computer Vision, Embedded Systems, Intelligent Applications

Languages:

Arabic (Native), English (Professional Working Proficiency)